ECON-7102: Environmental Economics I

Class 5: Instrument Choice (i)

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Today's Agenda

1. Questions?
2. Spillover from last class (intro. to policy instruments)
3. Instrument choice
   - Imperfect information
   - Weitzman (1974): Prices vs. Quantities
4. For next class...

Environmental policy with imperfect information

- Setting the stage
  - Economists tend to prefer **prices** to regulate externalities.
  - Regulators tend to prefer regulating the quantity of the pollutant.
- There is no theoretical reason to prefer taxes over quantities.
  - Prices = Pigouvian **taxes**
  - Quantities = **cap-**and-trade
- Central focus:
  - Policy is set with uncertainty about benefits and uncertainty about abatement costs.
  - **When should we prefer prices to quantities, and vice versa?**

Weitzman (1974) in a nutshell

- With uncertain abatement costs, you can either:
    1. Set **quantities** (be sure of $\bar{q}$), or
    2. Set **prices** (be sure of $\bar{p}$).

- The instrument you ***don't*** set will be uncertain.
    - If getting quantities wrong is very costly (e.g., nuclear waste leakage) $\rightarrow$ set $\bar{q}$.
    - If getting prices wrong is very costly (e.g., energy prices) $\rightarrow$ set $\bar{p}$.

Weitzman (1974)

Baseline framework

- There is some commodity q (e.g., emissions abatement) that can be produced with cost $C(q)$ and benefit $B(q)$.
- Assumptions: $B'' < 0$, $C'' > 0$, $B'(0) > C'(0)$, $B'(q) < C'(q)$ for large q.
- Planner maximizes net benefits:

$$\max_q B(q) - C(q) \quad \rightarrow \quad B'(q^\star) = C'(q^\star).$$

- Can implement solution by setting:
  - **price:** $p^\star = B'(q^\star) = C'(q^\star)$, or
  - **quantity:** $\bar{q} = q^\star$.

Weitzman (1974)

Introducing uncertainty

- Uncertain costs: $C(q, \theta)$ where θ is random variable.
- Uncertain benefits: $B(q, \eta)$ where η is random variable.
- Easiest to think of θ and η as information asymmetry (the regulator doesn't know true costs/benefits)

Weitzman (1974): Optimal **quantity** regulation

- Regulator maximizes expected net benefits:

$$\max_q \mathbb{E}[B(q,\eta) - C(q,\theta)]$$
$$\Rightarrow \quad \mathbb{E}[B_1(\hat{q},\eta)] = \mathbb{E}[C_1(\hat{q},\theta)].$$

- Choose $\hat{q}$ where expected MB = expected MC.

Weitzman (1974): Optimal **price** regulation

- A price delegates the quantity choice to the firm:

$$\max_{q} \; pq - C(q, \theta) \quad \Longrightarrow \quad C_1(q, \theta) = p$$

- The firm acts on its own knowledge of θ. Firm's best response is: $q = h(p, \theta)$.

Weitzman (1974): Optimal **price** regulation

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$$\max_{q}\ pq - C(q,\theta) \quad \Longrightarrow \quad C_1(q,\theta) = p$$

- The firm acts on its own knowledge of θ. Firm's best response is: $q = h(p,\theta)$.

- The regulator's best price supports the optimal expected quantity:

$$\tilde{p} \ \approx\ \mathbb{E}[C_1(\hat{q},\theta)] = \mathbb{E}[B_1(\hat{q},\eta)] \qquad \text{(to a first order...)}$$

(Exact condition: $\tilde{p} = \mathbb{E}[B_1(h(\tilde{p},\theta),\eta)\ h_1(\tilde{p},\theta)]\ /\ \mathbb{E}[h_1(\tilde{p},\theta)]$ – expected MB weighted by the firm's price response h_1.)

Weitzman (1974): Comparing prices and quantity regulation

- $\hat{q}$ fixes the quantity. $\tilde{p}$ fixes marginal cost. Both target the same $MB = MC$ condition.
- Neither $\hat{q}$ or $\tilde{p}$ will achieve an optimum ex post, so the question becomes:
  - Which type of regulation comes closer?
  - Under what circumstances?
- Define the difference between net benefits under the price ($\tilde{q}(\theta) \equiv h(\tilde{p}, \theta)$) and under the quantity $\hat{q}$:

$$\Delta = \mathbb{E}\big[(B(\tilde{q}(\theta), \eta) - C(\tilde{q}(\theta), \theta)) - (B(\hat{q}, \eta) - C(\hat{q}, \theta))\big]$$

$$\Delta = \text{(comparative advantage of prices over quantities)}$$

- After...
  - Some simplifying assumptions (Taylor approximations)
  - Substituting in FOCs
  - Assuming θ and η are mean zero and independent
- Then, Δ can be expressed more simply as...

Weitzman (1974): Main result and intuition

Relative advantage of prices over quantities:

$$\Delta \cong \frac{\sigma^2 B''}{2C''^2} + \frac{\sigma^2}{2C''}. \qquad \ldots(20)$$

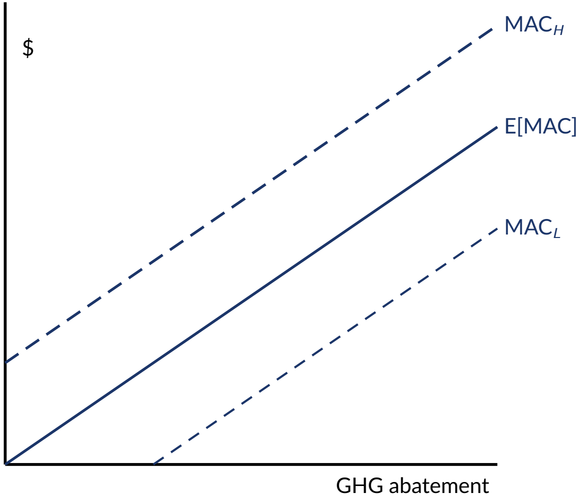
Expression (20) is the fundamental result of this paper.[1] The next section is devoted to examining it in detail.

[1] In the supply and demand context B'' is the slope of the (linear) demand curve, C'' is the slope of the (linear) supply curve, and σ^2 is the variance of vertical shifts in the supply curve.

- B'' and C'' are the slopes of marginal benefits / marginal cost curves
- σ^2 is mean-square error (i.e., uncertainty) of vertical shifts in supply curve
- Key results:
  - Sign of Δ is equal to the sign of $B'' + C''$. (Whether prices are preferred to quantity controls depends on the slopes of the marginal cost/benefits curves.)
  - When σ^2 (or uncertainty about costs) goes to zero, price and quantity controls are equivalent.
  - Uncertainty about *benefits* doesn't matter!

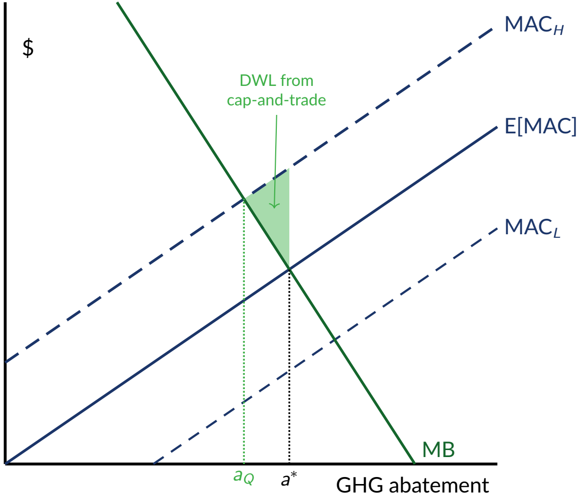
Prices vs. quantities, graphically (steep MB)

- MACs are unknown to the regulator.
- Expected MAC ($\mathbb{E}$[MAC]) is the regulator's best guess, but it could be higher (MAC_H) or lower (MAC_L).



Prices vs. quantities, graphically (steep MB)

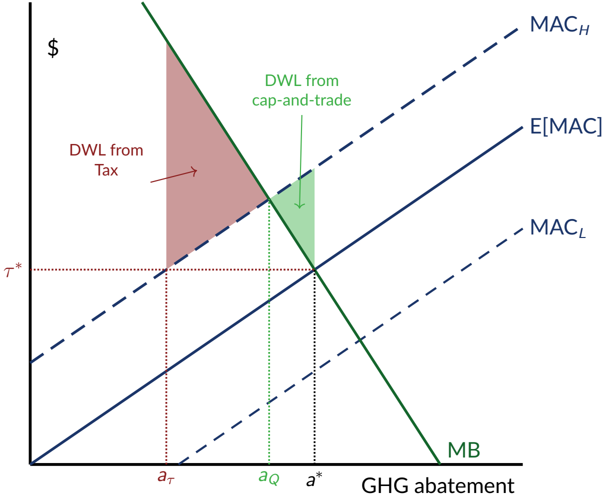
- *Imagine we have a "steep" MB curve, leading to optimal expected abatement, $a^\star$*
- The regulator chooses a **quantity instrument (i.e., cap-and-trade)** with cap at $a^\star$
- But, because costs are uncertain, we actually experience MAC_H
- If we had known MAC_H, we would have chosen a lower abatement level, a_Q, which is $< a^\star$.
- This results in DWL: We abated too much pollution relative to costs!



Prices vs. quantities, graphically (steep MB)

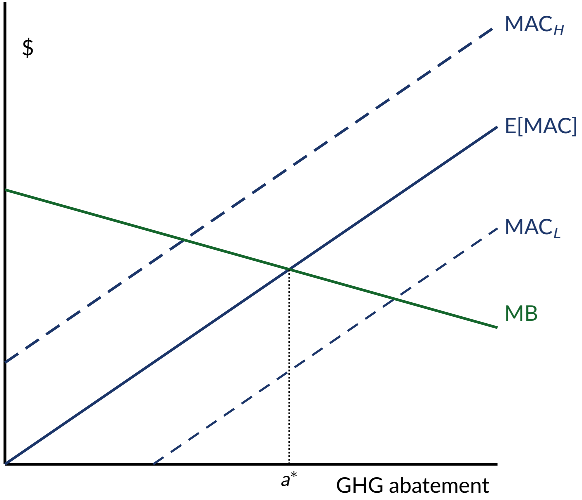
- *Imagine we have a "steep" MB curve, leading to optimal expected abatement, $a^\star$*
- Now consider a regulator that chooses a **tax** with $\tau^\star$ set to where $MB = \mathbb{E}[MAC]$.
- If we experience MAC_H, the tax generates even lower abatement than the quantity control and more DWL.

$\Longrightarrow$ with uncertain costs, quantity controls are preferred to a tax when MB is relatively steep.



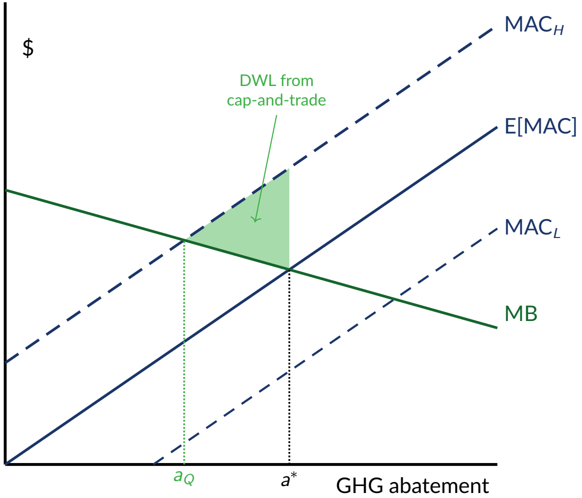
Prices vs. quantities, graphically (flat MB)

- Now, repeat this exercise with a relatively flat MB curve.



Prices vs. quantities, graphically (flat MB)

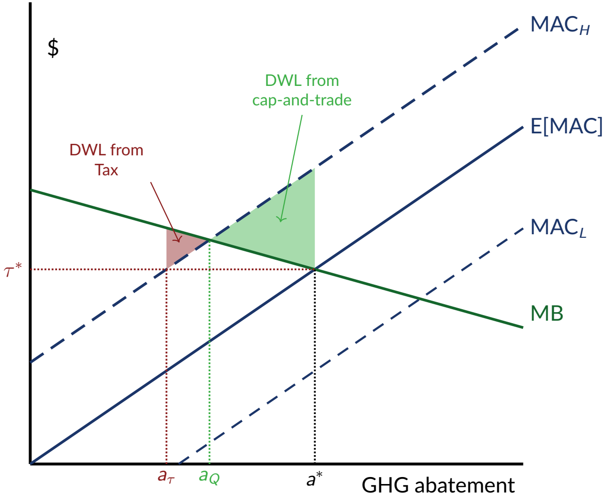
- Now, repeat this exercise with a relatively flat MB curve.
- Quantity regulation results in **green DWL triangle**.



Prices vs. quantities, graphically (flat MB)

- Now, repeat this exercise with a relatively flat MB curve.
- Quantity regulation results in **green DWL triangle**.
- Tax results in **red DWL triangle**.

$\Longrightarrow$ with uncertain costs, tax controls are preferred to quantity regulations when MB is relatively flat.



Key takeaways

- When costs are uncertain…
  - If we have relatively steep MB (e.g., slightly more emissions generate large increases in damages), then quantity controls are preferred.
    - This makes sense… we can better control the max quantity by setting a cap.
  - If we have relatively flat MB (marginal damage of an additional ton of CO_2), then price controls are preferred (e.g., like a carbon tax).

- Climate change tends to fall into the "flatter" MB world, where taxes are preferred.
  - But, there are other factors that matter (e.g., political feasibility).
  - Throughout the world, there is a mix of carbon prices (e.g., British Columbia) and cap-and-trade programs for GHGs (e.g., EU Emissions Trading Scheme, California cap-and-trade program).

Next week…

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- **First research abstract due Sunday** (we'll start class on Monday with a quick "pitch" in class).
- Focus on P&R readings (Ch. 5), but take a look at the papers as well.
- Goulder & Parry (2008) is a great, less-technical overview of textbook material.